

Application of the *Daubert* Standards to the Meyers Neuropsychological Battery Using the Rohling Interpretive Method

Martin L. Rohling¹ · John E. Meyers² · Gerard R. Williams³ · Stephen S. Kalat⁴ · Shanna K. Williams⁵ · Joshua Keene⁶

Received: 12 August 2015 / Accepted: 18 August 2015 / Published online: 8 September 2015
© Springer Science+Business Media New York 2015

Abstract The Meyers Neuropsychological Battery (MNB) is a neuropsychological assessment battery used to detect cognitive impairment caused by acquired brain injury. Results obtained by examinees given the MNB have been submitted as evidence in a judicial proceeding in cases involving traumatic brain injury (TBI) and other neurocognitive disorders. We provide an examination of the MNB when used with the Rohling Interpretive Method (RIM) through the lens of *Daubert v. Merrell Dow* (1993). *Daubert* established criteria to be applied by judges to determine who can provide expert opinions and what these experts are allowed to present to the triers of fact. *Daubert* has five criteria judges consider in their role as gatekeepers of an expert's testimony. These standards are utilized to ascertain if the expert's testimony is scientific,

with particular focus on its reliability, validity, and relevancy. We report on the MNB-RIM's ability to withstand the rigors of a *Daubert* analysis, with each criterion addressed in sequence. To accomplish this task, we review the peer-reviewed literature that has tested each of the MNB components, as well as the utility of the battery in its entirety. The literature extends over the 20-year history that the MNB and the RIM have been in use in clinical and forensic assessments. Statistics regarding the MNB-RIM's error rate have been empirically derived in numerous publications. These factors have led to a "general acceptance" of the battery and the material. Our review is intended to provide users of the MNB-RIM with the information they will need to successfully defend against a *Daubert* challenge.

✉ Martin L. Rohling
mrohling@southalabama.edu

John E. Meyers
Jmeyersneuro@yahoo.com

Gerard R. Williams
drgerardwilliams@charter.net

Stephen S. Kalat
drkalat@aol.com

Shanna K. Williams
shanna.k.williams@gmail.com

Joshua Keene
keenejosh22@gmail.com

Keywords *Daubert* · Neuropsychological assessment · Forensic neuropsychology · Rohling interpretive method · Meyers neuropsychological battery

There has been an increasing and widespread reliance on clinical neuropsychology in providing expert opinions in the courtroom (Greiffenstein & Kaufmann, 2012; Kaufmann & Greiffenstein, 2013; Sweet, King, Malina, Bergman, & Simmons 2002). Since the majority of licensed and practicing psychologists seem to be moving in the direction of evidence-based practice and were likely trained in either the scientist-practitioner model or the practitioner-scholar model, the field has been forced to develop assessment measures, tests, and batteries that are empirically supported (Sweet & Meyers, 2011). The expanding knowledge base generated from neuropsychological research informs "triers of fact," which includes judges, jurors, or mediators (Greiffenstein, 2008). Neuropsychologists are now, more than ever, required to develop extensive knowledge of various pathological conditions, as

- ¹ University of South Alabama, Mobile, AL, USA
- ² Comprehensive MedPsych Systems, Sarasota, FL, USA
- ³ Hurley Medical Center, Flint, MI, USA
- ⁴ Private Practice, Denver, CO, USA
- ⁵ Nova Southeastern University, Ft. Lauderdale, FL, USA
- ⁶ Delta Family Clinic, Flint, MI, USA

well as assessment and statistical interpretation skills, to adequately respond to increased scrutiny typically applied in a forensic setting. The ever expanding literature related to neuropsychological practice has paradoxically increased the degree of dueling amongst experts in the adversarial arena of forensic practice. Most neuropsychologists can find and cite published empirical research to support nearly any opinion they wish to offer. It is for this reason that the Daubert standards (Daubert v. Merrill Dow 1993) have been developed so sift through the evidence to separate the wheat from the chaff.

A long running argument in the field of clinical neuropsychology has been the “fixed” versus “flexible” battery debate. If anyone thinks this argument has been settled, refer to the recent New Hampshire Supreme Court ruling of *Baxter v. Temple*, which involved a child who was allegedly poisoned by lead paint at the age of 1 year old. The initial filing was in 2001 when the child was already 7 years old and the case settled in the fall of 2012, over a decade later and after the child had turned 18. One of the major disagreements in the case was the use of a flexible versus a fixed battery. Whether a test battery can be considered reliable and valid if it is fixed, flexible, or semi-flexible is based on empirical data that is published in peer-reviewed specialty journals (Sweet et al., 2002). To some degree, all batteries vary along a continuum defined at the poles as fixed and flexible. Examples of more fixed batteries, including the Halstead-Reitan Neuropsychological Battery (HRB; Reitan & Wolfson, 1985) and the Halstead Russell Neuropsychological System-Revised (HRNES-R; Russell & Starkey, 2001). Other batteries that allow for some degree of flexibility are the Neuropsychological Assessment Battery (NAB; Stern & White, 2003) and the WAIS-IV/WMS-III/ACS system (Wechsler, 2008). Regardless the degree of flexibility a battery’s authors’ allow, it is common that the fixed portion of the battery has undergone extensive testing. The research typically involves the assessment of hundreds of subjects, both normal and those who suffer from cognitive impairment believed to be caused by various neuropathologies.

More flexible batteries have gained wider acceptance, as described in *Baxter v. Temple* in an Amicus Brief submitted to the New Hampshire Supreme Court by the American Academy of Clinical Neuropsychology (2007). The AACN brief notes that 76 % of practicing clinical neuropsychologists utilizes a flexible battery approach. Only 7 % of clinicians reported that they relied on fixed batteries (Sweet, Meyers, Nelson, & Moberg 2011). The MNB contains both fixed and flexible components, as described by Meyers and colleagues (for examples, see Meyers, 2014; Meyers & Rohling, 2004; Meyers & Rohling, 2009).

While a flexible battery might rely on the validity that is inherent in its component tests, Rohling and colleagues have developed a systematic approach to flexible battery interpretation entitled the Rohling Interpretive Method (RIM; Miller

& Rohling, 2001; Rohling, Langhinrichsen-Rohling, & Miller 2003a; Rohling, Miller, & Langhinrichsen-Rohling 2004). We believe that the MNB, when analyzed via the RIM technique, has demonstrated superior proficiency in discriminating normal cognition from impairments associated with acquired brain injuries (Larrabee, Millis, & Meyers 2008; Larrabee 2012a, b; Rohling, Meyers, & Millis 2003b, Rohling, Williamson, Miller, & Adams 2003c). In reviewing the conflicts over the validity of test batteries, Kaufmann (2012) noted that every published legal case addressing this has found that a fixed battery offered no advantage over a flexible battery for forensic consultation (see also Larrabee et al. 2008; Rohling et al. 2003a).

The *Daubert* standard has defined two broad criteria for admissibility of scientific evidence in the courtroom. First, was the information that is to be submitted in court obtained in a scientifically valid manner? Second, are the results of an assessment relevant to the issue before the court? Other guidelines have been put forth to better delineate what is considered valid and what is considered relevant. An abbreviated listing of the criteria adopted by *Daubert* is presented in Table 1. As noted by Greiffenstein and Kaufmann (2012), these criteria are unweighted, polythetic, and non-exclusive. “This means no single element is necessary and any single element or combination of elements is sufficient to admit into evidence” (p. 51). Therefore, none of the factors alone are dispositive. The party offering an expert opinion bears the burden of demonstrating that the conclusions offered are based on sound science (Russell, 2012). Thus, the examiner must show that the battery included standardized tests that have been subjected to peer-review and have been shown to be valid.

The current paper is designed to review the historical developments that have led up to the current version of the MNB, which includes its use of the RIM interpretive system, and to see if it meets the *Daubert* standards. If so, results from a MNB-RIM assessment should be admitted into evidence, via expert testimony, to assist the court in case adjudication.

Developments of the MNB and RIM

Approximately 20 years ago, Meyers began working on developing a battery of tests (Meyers & Diep, 2000) that is now called the MNB (Meyers, 2014). Initially, the Meyers’ battery consisted of several commonly administered tests that came from other batteries and were well-known to the neuropsychological community. The tests selected came from a variety of existing batteries, including the Halstead Retain Battery (HRB; Reitan & Wolfson 1985), Benton’s Iowa Neuropsychological Battery (Benton, Savin, de Hamsher, Varney, & Spreen 1994), and the Boston Diagnostic Aphasia Exam (BDAE; Kaplan, Goodglass, & Weintraub 1983). The tests selected for inclusion in the MNB by Meyers had been listed

Table 1 The standards listed below are used during a *Daubert* challenge, which is designed to determine if an expert's opinion should be admitted into testimony. The issue at hand is whether the information an expert is planning on presenting in court should be heard by the jury

Item	Criteria
1.	Has the MNB been tested scientifically for accuracy?
2.	Has the MNB been subjected to peer-review and publication?
3.	Does the MNB have a known error rate (reliability and validity)?
4.	Has the MNB been generally accepted by the scientific community and mainstream clinical neuropsychologist?
5.	Does the MNB have a method of controlling its use?

in various compendiums devoted to neuropsychological instruments and norms (i.e., Mitrushina, Boone, & D'Elia 1999; Spreen & Strauss, 1991, 1998; Lezak, 1995). The frequency of use for these instruments has since been documented by Rabin, Barr, & Burton (2005). It is now known that the instruments chosen for inclusion in the MNB constituted over 50 % of the tests ranked in the top 30 frequently used measures by neuropsychologists.

The battery was administered to both patients and control subjects. Control subjects were hospitalized patients who had no history of acquired brain injury, but had been admitted to undergo various other medical procedures and who volunteered for an assessment to assist in the development of the MNB. The data obtained from these assessments were analyzed via to identify which had the highest level of sensitivity and specificity in various pathological groups. The results were surprising, as only four or five test scores were needed to discriminate most clinical entities from normal subjects. Additional tests were required to discriminate scores obtained by patients who suffered from various types of neuropathology from those obtained from patients who had suffered a TBI. By determining which tests were the most sensitive to the neuropathology, the battery was whittled down to its current state and was initially named the Meyers Short Battery (MSB). The MSB was subsequently renamed the MNB, so as to avoid confusing examiners that it was to be used for screening purposes only.

Again, some of the criteria used to select tests for inclusion in the battery were: (1) How available was a test to acquire for those interested in using the battery. Tests that were “in the public domain” (i.e., without copyright) were preferred over those that were owned by publishers and expensive to acquire. (2) How well known was a test to the neuropsychological community. Tests that had a large number of professionals who were familiar with it were selected over those that were more obscure. (3) How frequently did members of the profession administer the test. Those that were more frequently given were selected over those that were less commonly given. Finally, (4) how easy was a test to administer and score. Those that were simpler to give and score were selected over those that had several nuances. For example, the Wisconsin

Card Sorting Test (WCST; Heaton, Chelune, Talley, Kay, & Curtiss, 1993) performed approximately as well as the Category Test (CT; Reitan & Wolfson, 1985; Labreche, 1983) upon initial screening for selection. These two tests have often been assumed to measure executive functioning, which was the purpose of examining them to begin with. The CT was retained as the preferred measure of abstract reasoning and conceptual processing over the WCST because it is easier to administer, score, and interpret than is the WCST. Subsequently, it has been shown that the two tests do not measure the same construct (Perrine, 1993), nor frontal lobe functioning (Demakis, 2004). Nevertheless, this was the logic used for selecting the CT.

The MNB can typically be administered and scored in approximately 3 h. The scores obtained are assigned to various cognitive domains, i.e., verbal reasoning, visual reasoning, verbal memory, visual memory, performance validity, and emotional/personality. The assignment of scores to domains is based on results from various factor analytic studies that have appeared in the literature (e.g., Larrabee et al. 2008; Larrabee, 2000; Leonberger, Nicks, Larrabee, & Goldfader 1992). Table 2 provides a description of the core MNB tests, which are presented in the order of their administration. Test order was organized by Meyers (2014) to prevent “overlap,” such as two verbal lists running congruently. Although the battery is generally given in the specified order, small variations in administration have not been shown to adversely affect influence obtained scores (Meyers, 2014).

At the same time that Meyers was in the process of developing his battery, Rohling began his work on developing an interpretive system designed to for use with individual case data (Miller & Rohling 1998, 2001; Rohling, Miller, & Langhinrichsen-Rohling 1998). In the study of Miller and Rohling (2001), the purpose for developing the RIM was stated as follows:

Williams (1997) recommended the development and use of technology that would function as a decision aid. These decision aids would help neuropsychologists correctly interpret psychometric test data and increase the accuracy of their diagnoses (Sicol, 1989). Meehl (1954) was the first to advocate for additional reliance

Table 2 Basic information on the MNB in administration order

Test #	MNB test	Original norms	Assigned domain
1.	Wechsler scale (adult and child); information, similarities; block design ^a ; picture completion; arithmetic, digit span ^a ; digit symbol (coding) ^a	Wechsler (1997, 2003, 2008); Ward (1990); Pilgrim et al. (1999); Meyers et al. (2013b)	Designated in manuals Verbal reasoning Visual reasoning Attention processing Speed
2.	Forced choice (FC) ^a	Brandt et al. (1985)	Attention
3.	Rey complex figure test (RCFT)—Copy ^a	Meyers & Meyers (1995a, b)	Visual reasoning
4.	Animal naming	Spreeen & Strauss (1998), p. 457, Tables 11–12, p. 458, Tables 11–14	Attention
5.	1 min estimation	Lezak et al. (2004), p. 636; Meyers (2014)	Attention
6.	3 min recall ^a of RCFT	Meyers & Meyers (1995a, b)	Visual memory
7.	Controlled oral word association test (COWA)	Spreeen & Strauss (1998), p. 455, Tables 1–9, p. 458, Tables 11–13	Verbal reasoning
8.	Dichotic listening—left, right, both ^a	Roberts et al. (1994); Meyers et al. (2002b)	Verbal reasoning Processing speed
9.	North American Adult Reading Test	Spreeen & Strauss (1998); Meyers (2014)	Premorbid estimate
10.	Sentence repetition	Spreeen & Strauss (1998), pp. 370–371, Tables 10–25, 10–27; Read & Spreeen (1986); Meyers et al. (2000)	Attention
11.	RCFT 30-min Recall ^a , Recognition Trial ^a	Meyers & Meyers (1995a, b)	Visual memory
12.	Rey Auditory Verbal Learning Test (AVLT)	Spreeen & Strauss (1998), pp. 333–335 Tables 10–16 to 10–18; Ivnik et al. (1990)	Verbal memory
13.	Judgment of Line Orientation (JLO ^a)	Benton et al. (1994), pp. 50–53, Tables 5–3 & 5–6.	Visual reasoning
14.	Boston Naming Test (BNT)	Spreeen & Strauss (1998), pp. 436–438 Table 11–5 and 11–2	Verbal reasoning
15.	Finger Tapping (FT)—dominant hand ^a and non-dominant hand	Reitan & Wolfson (1985); Heaton et al. (1991, 2004)	Motor
16.	Finger Localization (FL)—dominant ^a and non-dominant	Benton et al. (1994); Spreeen & Strauss (1998), p. 558, Table 13–3	Sensory
17.	Trail Making Test Parts A & B (TrA & TrB)	Reitan & Wolfson (1985); Heaton et al. (1991, 2004)	Processing speed
18.	Token Test ^a (TT)	Spreeen & Strauss (1998), p. 475, Tables 11–16, 11–17	Verbal reasoning
19.0	AVLT 30-min recall and recognition ^a	Spreeen & Strauss (1998), pp. 333–335, Tables 10–16 to 10–18; Ivnik et al. (1990)	Verbal memory
20.	Booklet Category Test (Victoria Revision)	Spreeen & Strauss (1991); Kozel & Meyers (1998); Heaton et al. (1991, 2004)	Visual reasoning

^a Test that is a part of a performance validity measure

on actuarial data for decision making in psychological assessment... Although common judgment errors can threaten seriously the reliability and validity of multiple test battery interpretation (Dawes, Faust, & Meehl 1989), we believe that neuropsychologists who combine their statistical training with technology that is readily available (e.g., personal computers and statistical software) can address effectively these biases. However, failure to do so reduces the probability of successfully discriminating between real cognitive loss and change and chance variability in the scores.

Miller and Rohling (2001) further noted that the RIM is well grounded and statistically rigorous, stating:

The methodology used in this process is similar to that recommended for meta-analytic reviews of research literature (e.g., see Glass, McGraw, & Smith 1981; Rosenthal, 1984; Rosenthal, Rosnow, & Rubin 2000). The linear combination of scores we recommend is supported by the research of Dawes (1979) and Dawes and Corrigan (1974) and follows a similar model as presented by Kiernan and Matthews (1976). It is a statistically sound method of analysis that produces summary results using a flexible battery that are analogous to those generated in fixed battery approaches, including the Halstead Impairment Index (HII) of Reitan and Wolfson (1985), the Average Impairment Rating (AIR) of Russell, Neuringer, & Goldstein (1970), and the Global

Neuropsychological Deficit Scale (GNDS) of Reitan and Wolfson (1988). Thus, a primary advantage of the RIM, in comparison with other indices, is that it does not restrict a clinical neuropsychologist to a particular battery of tests (pp. 144–145).

During the first 6 years of the development of both the MNB and the RIM, Meyers and Rohling had not been acquainted, despite the fact that they lived within close geographic proximity of each other, living and working just 150 miles from one another (Sioux City, IA, & Lincoln, NE, respectively). However, they conducted their research without the knowledge of one another's work. Rohling wished to create a system of individual test score analysis that is based on meta-analytic procedures typically used for large sample group analyses. The concept of the MNB was created by Meyers when he became frustrated with assessments that often required patients to endure up to 8 h of testing. He wanted to determine how many test scores were necessary to provide reasonably accurate diagnostic conclusions for most cases.

In the remainder of this paper, we will detail the MNB-RIM's ability to meet or exceed each of the five *Daubert* standards. *Daubert* criteria will be presented in question and answer format, which will allow readers to consider each criteria applied to the MNB-RIM from a critical perspective. This skeptical view places the burden of proof for admissibility of the results from an MNB-RIM assessment into evidence and will assist users in responding to legal challenges when administering an MNB-RIM assessment for forensic purposes.

The Five Criteria of *Daubert*

Criterion 1: Has the MNB-RIM Been Tested Scientifically for Accuracy?

Yes. First, as noted above, the MNB was developed by selecting tests for the core battery based, in part, on their established reliability and validity. Each test in the core battery might be capable of independently passing the *Daubert* criteria. The MNB tests use methods that are replicable and result in diagnostic conclusions that are falsifiable. Individual users can flexibly add tests. However, adding tests does not alter the results provided to users from the core battery. Tests are added to augment the assessment and are based on individual users' personal preference.

Criteria 2: Has the MNB-RIM Been Subjected to Peer-Review and Publication?

Yes. Although Meyers and Rohling developed their systems using rigorous and exhaustive methods, both the MNB and

the RIM have continued to be refined over time. In fact, several researchers have published peer-reviewed manuscripts on the properties of both systems. For example, using Google Scholar as a search engine, over 40 researchers were identified as having examined aspects of the MNB and over 30 researchers have investigated aspects of the RIM. Using Google as a search engine and entering in quotations the "Meyers Neuropsychological Battery" resulted in 804 hits from the Internet. Using the same process, but entering the "Rohling Interpretive Method" resulted in 760 hits. When a primary outcome measure of the RIM, the "Overall Test Battery Mean," was entered in quotation marks, it resulted 1360 hits. Finally, as of mid-2015, Meyers has published 52 manuscripts and abstracts that have been cited 2065 times. Rohling has published 89 manuscripts and abstracts that have been cited 3200 times (Publish or Perish; <http://www.harzing.com/>).

When examining the association between year of publication and citations, all regression equations for both Meyers and Rohling, whether based on publications or citations, were positively sloped, indicating both that the MNB and the RIM are experiencing growth in their examinations by independent researchers, as well as garnering increased interest by clinicians. In total, 65 manuscripts have been published relating to the two systems. Meyers and/or Rohling have authored 40 of these manuscripts. However, the two have jointly published only 4 papers together. Thus, 25 manuscripts have been authored by various independent researchers (e.g., Brad Axelrod, Russell Bauer, Stephen Bowden, Jacobus Donders, Paul Green, Michael Kirkwood, Glenn Larrabee, Robert McCaffrey, & Scott Millis).

The MNB and/or the RIM have been examined in at least 65 publications. Such an extensive literature supports the idea that the two systems are widely used by neuropsychologists, and their use has grown substantially over the past decade. The widespread use of the MNB-RIM lends itself to further development regarding its efficacy, effectiveness, and efficiency. Miller, Fichtenberg, & Millis (2010) in their review noted:

... Meyers and Rohling (2004) have made a major contribution to the ability focused battery (AFB) by conducting validation studies of the Meyers Neuropsychological Battery (MNB), an example of the AFB that has been in clinical use for years... The MNB has demonstrated that a flexible core approach is capable of performing in a manner comparable to a fixed battery... The MNB has also demonstrated sensitivity to different causes of cognitive dysfunction. (p. 679)

Carone and Bush also commented on the MNB-RIM in their edited text devoted to the issue of mild traumatic brain injury (2013, p. 285 & pp. 288–289). These researchers described it as using comparison groups and discriminant

functions to identify various diagnostic groups. The MNB-RIM was positively reviewed by Bush (2011) as well in the *Encyclopedia of Clinical Neuropsychology* (pp. 1589–1590). Russell (2012) discussed the characteristics of the MNB-RIM and noted the consistently high levels of diagnostic accuracy, with a sensitivity of 93 %, specificity of 90 %, and overall predictive power of 92 %. Russell concluded that “Undoubtedly, the MNB will remain an excellent usable standardized battery (p. 298).”

Criteria 3: Does the MNB-RIM Have Known Error Rates (i.e., reliability and validity)?

Yes. This criterion addresses the MNB-RIM’s reliability and validity in the detection of cognitive impairment as it relates to acquired brain injury. To produce a known error rate, it is important to establish internal and external validity when measuring the capabilities of a psychological test instrument. It is necessary for examiners to control for the influence of spurious variables when conducting a test procedure to establish internal validity. Controlling for outside influences is necessary to establish a battery’s reliability. There must also be a high degree of weight placed on external validity. Results should be consistent in its results across comparable populations.

Reliability A primary objective of Rohling during the development of the RIM was to increase the inter-rater reliability of diagnostic outcomes of neuropsychological assessment, particularly in the forensic setting. Prominent psychological researchers have identified common heuristics frequently used when making diagnostic decisions that result in errors (e.g., Tversky & Kahneman, 1974). In fact, to address these errors generated when using a flexible or semi-fixed battery, Miller and Rohling (2001) noted:

... The risks involved with the evaluation and interpretation of multiple measures are many and tend to increase at a rate even greater than the number of tests used (Ingraham & Aikken, 1996). These risks include inflated error rates, multicollinearity, weighting decision problems, unknown veracity/reliability of sets of tasks, variability across interpreters, and a number of human judgment errors (Wedding & Faust, 1989). (Miller & Rohling, p. 143) Thus, Rohling and colleagues set out on their development of the RIM. They worked to improve the reliability of diagnostic outcomes by clinical neuropsychologists by developing a system that reduces bias and increases accuracy.

The reliability of the MNB-RIM has been studied using a general clinical sample of 63 subjects. These individuals were tested twice, an average of 19.1 months between testing

sessions. The MNB-RIM obtained an overall reliability for the OTBM of between .86 and .93 (Meyers & Rohling, 2004; Rohling, Miller, Axelrod, Wall, Lee, & Kinikini 2015). The reliabilities of the various subtests ranged from .46 (Rey Auditory Verbal Learning Test - Trial 1) to .88 (WAIS-III Block Design) with a mean of .72.

Furthermore, the MNB-RIM uses commonly administered tests that are presented in a standard order. The MNB-RIM’s battery has built in rest breaks to ensure that fatigue is not a factor that negatively influences performance. The test administration order was carefully designed to prevent “overlap,” such as two verbal learning tasks are administered consecutively or concurrently. The order of test administration is one of the methods used for maintaining control over the use of the MNB-RIM.

Validity—Internal Internal validity is defined as the degree to which an independent variable causes a change in the dependent variable (Goodwin, 2010). Tests must carefully control for spurious influences created by confounding variables to determine a causal relationship between the variables of interest. The MNB-RIM uses well-established and empirically supported tests with a high degree of internal consistency and validity (Strauss, Sherman, & Spreen 2006). These tests use repeatable administrative procedures to produce internally valid results. A newly developed standard of practice in test batteries for neuropsychological assessment is the inclusion of validity measures (Bush, Ruff, Tröster, Barth, Koffler, Pliskin, Reynolds, & Silver 2005; Heilbronner, Sweet, Morgan, Larrabee, Millis, & Conference participants 2009). The MNB-RIM contains 11 internal PVMs (Meyers, 2014), which are listed in Table 2 and are identified with an asterisk (*). Each of these has been included through the establishment of their accuracy based on peer-reviewed published research.

Using a combination of nine PVMs, the MNB-RIM had an invalid response rate of 36 % for a sample of known subjects (Meyers & Volbrecht, 2003). This base rate is quite similar to the 40 % reported by Larrabee, Millis, & Meyers (2009). Using data from identified groups (non-litigating: LOC < 1 h, LOC > 1 h, LOC 1–8 days, LOC 9+ days, pain, depression, normal, litigating: LOC < 1 h, LOC > 1 h, pain and malingering actors), the PVMs had a sensitivity to invalid responding of .83, a specificity of 1.00, and an overall correct classification rate of 95 %. These nine PVMs may be more conservative than other freestanding measures (e.g., Word Memory Test or WMT; Green, 2007). Nevertheless, Meyers, Volbrecht, Axelrod, & Reinsch-Boothby (2011b) found that the PVMs accounted for a similar amount of variance in neuropsychological test scores as did the WMT, which was found to be 52 % (Green, Rohling, Lees-Haley, & Allen 2001) and the Test of Malingered Memory (TOMM; Tombaugh, 1996), which has been reported to

be 47 % (Constantinou, Bauer, Ashendorf, Fisher, & McCaffrey 2005). Nearly identical results were found in a German sample with the WMT and the Medical Symptom Validity Test (MSVT; Green, 2004), with 45 % of the variance accounted for by scores in the performance validity tests (Stevens, Friedel, Mehren, & Merten 2008).

A new PVM that was added to the MNB-RIM since the publication of the original nine PVMs is the WMTMNB algorithm. This new PVM was made using a sample of 264 consecutive referrals, which were referred for a neuropsychological assessment, and were given the MNB and the Word Memory Test (WMT; Green, 2003; Green, 2007). All were identified as either passing or failing the WMT, based on standard interpretation and administration, using both cutoff scores and genuine memory impairment patterns. All the MNB-RIM variables that are not part of the PVM already described were included in a stepwise discriminant function (Pohar, Blas, & Turk 2004; Maroco, Silva, Rodriquez, Guerreiro, Santana, & de Mendonça 2011). The discriminant did not include any of the WMT scores, but rather predicted results from the WMT, based on scores below the cut score and the Genuine Memory Impairment Profile (Green, Flaro, & Courtney 2009). A score below 0 is a fail on this PVM. The discriminant function was able to correctly classify 70 % of the sample. To continue with the setting of false-positive rates at zero, for patients with less than 7 days of loss of consciousness (LOC), a score of $-.50$ is recommended.

It is recommended that the MNB-RIM be administered with the Minnesota Multiphasic Personality Test-2 (MMPI-2; Butcher, Dahlstrom, Graham, Tellegen, & Kaemmer 1989) or the Minnesota Multiphasic Personality Test-2-Reformatted (MMPI-2-RF; Tellegen & Ben-Porath, 2008). When examiners administer one of the two versions of this personality test, then the Meyers Index (MI; Meyers, Millis, & Volkert 2002a; Meyers, Miller, Haws, Murphy-Tafti, Curtis, Rupp, Smart, & Thompson 2013a, Meyers, Zellinger, Kockler, Wagner, & Miller 2013b) is available to assess the validity of the examinee's responses to emotional/personality questions.

In a recent study by Meyers et al. (2014a, b), the now 11 PVMs were studied to determine the sensitivity and specificity of failing any combination of these measures. Thus, a probability of the test scores obtained by a patient of being invalid can be objectively calculated by chaining likelihood ratios, which dramatically improves detection rates of invalid responding, with corresponding increases in the sensitivity and specificity, based on a post-test odds ratios of between .934 and .999. Additionally, the MNB-RIM manual presents PVM data at the domain level so that validity of performance at this level can be assessed. The same data were used by Meyers et al. (2013a, b) to determine the probabilities that

are associated with domains-level PVMs and inter-individual variability.

Validity—External In the *Standards for Educational and Psychological Testing* (AERA/APA/NCME, 2014), external validity is defined by the degree to which results can be generalized to various populations, settings, and confounding variables. Evidence used to establish external validity depends on a combination of patient history, initial assessment conducted by a qualified psychologist, presenting problems as divulged by the patient, observations by the therapist, the patient legal history, other self-report measures obtained from the patient, and brain imaging techniques used by qualified professionals. All of these factors contribute to the establishment of external validity. There is evidence comparing MNB-RIM results with these factors, which support the external validity of the battery. Meyers, Volbrecht, & Kaster-Bundgaard (1999b) found 94.4 % accuracy in distinguishing between individuals capable of driving and those advised by their physician not to drive. There was 84 % agreement between positive CT and MRI brain scans (Meyers & Rohling 2009) and scores on the MNB-RIM. The subjects who were tested twice had an average 19.1 months between sessions. The test–retest reliability for the two administrations was .86 (Meyers & Rohling, 2004). The length of loss of consciousness (LOC) was highly related to the severity of cognitive impairment, as measured by the MNB-RIM (Rohling et al. 2003b). Further study of the MNB-RIM was undertaken to identify its sensitivity to cognitive impairment. Rohling et al. (2003b) found in a sample of 436 TBI patients that the length of loss of consciousness (LOC) was related to the degree of cognitive impairment, as measured by the overall test battery mean (OTBM). Results were nearly identical to those presented by Dikmen, Machamer, Winn, & Temkin (1995), who used the expanded Halstead Reitan Battery (HRB) with group composite measures having a Pearson correlation coefficient of .98 (Rohling et al. 2003a, b, c). An OTBM, generated from HRB data, correlated .87 with the General Neuropsychological Deficit Scale (GNDS; Reitan & Wolfson, 1997). It correlated .90 with the Average Impairment Rating (AIR; Russell et al. 1970). Finally, it correlated .79 with the Halstead Impairment Rating (HII; Reitan & Wolfson, 1985). It was sensitive to neurological impairment in the sample of .90, a specificity of .32, a positive predictive value of .70, a negative predictive value of .65, and an overall correct classification rate of 69 %. The overall correct classification rate is consistent with that obtained from the traditional HRB summary scores (GNDS=71 %; AIR=65 %; HII=65 %).

An OTBM from the MNB-RIM was sensitive to cognitive impairment caused by other neurological and psychiatric conditions, with an overall sensitivity of .90, specificity of .99, and an overall correct classification rate of 96 %. These results indicate that the MNB-RIM was as sensitive to the degree of

cognitive impairment as is the more time-consuming Halstead Reitan battery. Examinees who are administered the HRB spend nearly 10 h being assessed compared to 3 h when given the MNB-RIM.

The MNB-RIM has been studied in relation to test performance by referring plaintiff and defense attorneys. In a study by Meyers, Reinsch-Boothby, Miller, Rohling, & Axelrod (2011a), the assessment results from 48 plaintiff referrals and 63 defense referrals were examined. Results showed that there were statistically reliable differences in the scores obtained on the performance validity measures. Thus, the MNB-RIM is not considered to be overly stressful for either plaintiff or defense referrals.

Norms The process of developing norms is detailed in the MNB-RIM manual (Meyers, 2014). Examination of Table 2 lists the references for the original norms. In evaluating the original normative data, it was found that there were variations in test norms by age and education. For example, the AVLT normative data from Spreen & Strauss (1998; p. 335) showed a mean of 11.4 (SD=2.4) for trial 6 (immediate recall) for subjects in the 30–39 years of age group. However, if the examinees were 40 years of age, the mean listed is 10.4 (SD=2.7). This means that if an individual scored 10 on this test 1 day before a birthday and was (theoretically) tested on the birthday, the T score associated with the raw score improves from 44 to 48. Using the Heaton, Grant, & Matthews (1991) normative system, a raw score of 10 would go from being in the below average category to the average category simply by growing 1 day older. This is a common problem with non-smoothed normative data. Another concern was the effects of various demographic variables (i.e., education, gender, handedness, and ethnicity), which might not have been significant in a normal population, but becomes a factor in a pathological sample. Thus, the MNB norms are generated from patients who were referred for testing out of concern that they suffered from a neurological or psychiatric condition. In response to these concerns, Meyers decided to smooth the data based on subjects' demographics. This was done by first selecting all subjects from the larger data set who earned a validity score of either 0 or 1 (failures) and were 15 years of age or older. The age limit was selected based on the age limit developed for the Trail Making Test by Reitan and Wolfson (1985). It was retained to maintain historical consistency. The total sample included 1727 patients with a mean age of 45.7 (SD=20.7) and a mean level of education of 12.3 years (SD=2.7). The sample consisted of 779 females and 948 males. Of these, 1543 were right handed and 184 were left handed. In terms of ethnicity, there

were 1617 White non-Hispanics, 32 mixed backgrounds, 27 Native Americans, 27 Latino Americans, and 2 Asian Americans.

Regression equations were generated using raw scores and the demographic variables of age, education, gender, ethnicity, and handedness to predict T scores previously calculated using the standard normative data from Heaton et al. (1991). Once the regression equations were calculated, they were used to generate regression-based T scores for each test. It was found that this procedure worked well for all test scores except the Token Test (adult) due to its skewed distribution properties. For the Token Test, percentile scores were calculated and converted to T score equivalents.

By examining the normative data of the MNB, it is apparent that the regression equations adequately predicted the T scores calculated using the original norms ($r=.90$). The purpose of smoothing norms is to correct for “table jump.” T scores for either adults or children can be calculated. The addition of demographic variables in the regression equations adjusts for differences that are known to influence test performance. The published literature that has examined the MNB has used these regression-based norms in the analyses.

Rohling et al. (2015) conducted a comparison of the MNB-RIM normative system with two other normative systems to see what similarities and differences might exist, as well as to try and determine which system gave the most accurate results. Using a random sample of 330 randomly selected referred cases, as well as 99 undergraduate volunteers, they compared normative scores from the MNB-RIM, Heaton's comprehensive scoring system (Heaton, Miller, Taylor, & Grant 2004) and Mitrushina, Boone, Razani, & D'Elia (2005) normative systems, which was generated using meta-analytic regression techniques (MA-Reg). All three systems were highly correlated with one another, with the average correlation for the Overall Test Battery Mean (OTBM) being .96. The authors concluded that all three systems were generally acceptable and, for the most part, congruent with one another. However, without a gold standard upon which to base a judgment, their tentative conclusion was that “... the Meyers system is the preferred system...” In making this judgment, they took into consideration the similarities and difference amongst the systems, as well as how well each system accounted for the error variance associated with the demographic variables of age, education, ethnicity, and gender. These confounds can be statistically corrected for, but the risk of applying such corrections is challenging, as one might over or under correction based on limited sampling of the variable of concern, in combination with the other variables of concern. Despite the fact that the normative sample developed by Meyers has very few minority subjects included, the relationships amongst the three normative systems were actually greater in magnitude for the subset of subjects who were

considered to be members of a demographic minority than they were for White subjects.

Another study by Meyers and Rohling (2009) identified the associations amongst scores from various tests and findings of brain lesions based on neuroimaging results (i.e., CT scans and/or MRI data). Test scores were examined from a sample of 124 subjects who had a positive CT or MRI findings located in discrete locations. The results showed that the tests that make up the MNB-RIM are significantly correlated with CT and MRI results, and scores were depressed when subjects had lesions in the brain areas that were considered a priori to influence performance on such tasks. The MNB-RIM scores were in agreement with the location of lesion 84 % of the time. The MNB-RIM is not only sensitive to the presence of brain lesions, but the magnitude of the lesions as well. The MNB-RIM also possesses ecologic validity, as it has been shown to be sensitive to complex tasks such as driving (Meyers et al. 1999b).

Criteria 4: Has the MNB-RIM Been Generally Accepted by the Scientific Community and Mainstream Clinical Neuropsychologists?

Yes. General acceptance can be demonstrated through multiple methods. First, the description and review of the component tests exist in commonly referenced compendiums (e.g., Mitrushina et al. 2005; Strauss et al. 2006) and the large number of articles published on the MNB-RIM. The references for this article are only a subset of the manuscripts that have been published regarding the MNB-RIM.

Acceptance has also been indicated by favorable reviews of the MNB-RIM in the peer-reviewed journals: Miller et al. (2010), whose review was cited above. Larrabee (2012a, b) noted the findings of Rohling et al. (2003a, b, c) related to a flexible battery employing a core set of tests in the various domains found in the MNB-RIM showed the same dose–response association with head injury severity found by Dikmen et al. (1995) in their seminal prospective study of neuropsychological outcome in traumatic brain injury. “What was particularly impressive is that the Overall Test Battery Mean (OTBM) average T score for the Meyers Neuropsychological Battery (a core for a flexible battery) was 39.2, essentially identical to the expanded HRB OTBM of 38.9; additionally, the MNB-RIM correlated .99 with the five levels of TBI severity...” Further, the OTBM from the MNB-RIM and the Halstead Reitan “correlated .97 with one another across the five levels of trauma severity (p. 14).”

General acceptance can be inferred by the growth of the MNB-RIM e-mail user group, which now is over 360 members. Another method is based on the number of computers registered to run the MNB-RIM software, which has grown to include nearly 1000 neuropsychologists. These data suggest that the number of MNB-RIM users now exceeds the number

of users estimated by Sweet et al. (2011) of the HRB (Reitan & Wolfson 1985) and HRNES-R (Russell & Starkey, 2001). The number of MNB-RIM users has steadily grown by 35 % per year. The annual MNB-RIM workshop series is in its 8th year. This is a 2-day event designed to train new as well as experienced MNB-RIM users in the latest findings published in the literature. The initial workshop was attended by approximately 25 neuropsychologists. The past two workshops have grown to just under 100 neuropsychologists.

Criteria 5: Does the MNB-RIM Have a Method Of Controlling its Use?

Yes, there are methods for controlling the use and users of the MNB-RIM. More specifically, this criterion addresses the questions: (a) Who can administer and interpret the test battery? (b) Are the procedures for administration and interpretation well standardized? (c) Are these details described in a published manual?

The manual establishes and maintains the standards that control the MNB-RIM’s administration, scoring, and interpretation. The materials sold are restricted to qualified professionals, which can be defined as licensed psychologists. The MNB-RIM manual, which is provided to users when they purchase the software, details (a) the tests included in the core battery, (b) the process of developing norms, (c) the normative data, (d) the order of test administration, (e) the incorporation of the RIM in the diagnostic process, and (f) the standard use of performance validity indicators. In addition, when users purchase the MNB-RIM package, they are provided with materials needed to administer the tests (e.g., picture of the right and left hand for the finger localization test; wooden blocks for the Token Test, etc.) The manual controls the ongoing development and additions to the MNB-RIM in the future.

Detailed instructions for the administration of each test included in the MNB-RIM are provided in the MNB-RIM test protocol. The protocol functions as instructions for the administration of the battery. It includes specific areas for the recording the responses provided by examinees. Examiners use these testing procedures to ensure that the MNB-RIM is given in a standardized manner. These procedures are also described in detail in the MNB-RIM user manual.

The guidelines for interpreting MNB-RIM data are also outlined in detail in the manual. Quantitative data are used by the examiner to develop treatment plans that are more likely to lead to improved outcomes. The profile matching approach allows clinicians to remain more objective during the interpretation phase of assessment. As a result, clinicians are more likely to avoid arbitrary classification or simplistic dichotomies (e.g., impaired vs. normal).

Besides the profile matching interpretation guidelines (Maroco et al. 2011), as noted previously, the MNB is compatible with the Rohling Interpretive Method (Rohling et al.

2003a, c, 2004). Contained in the MNB-RIM software is an artificial intelligence in the form of an Artificial Neural Network (Maroco et al. 2011; Meyers, Miller, & Tuita 2014b), which is used to improve the objectivity and accuracy of the profile matching and interpretation. Criterion 2 addressed the large body of peer-reviewed literature citing the MNB-RIM interpretation process. The use of computer models reduces human error and lends a high degree of inference to the results of the MNB-RIM. In this way, the RIM maximizes the utility of the MNB. Computer models are also able to quickly simplify and organize information into easily interpretable results, thus reducing potential errors made by practitioners (Dawes et al. 1989).

Summary and Conclusion

The MNB-RIM meets or exceeds the criteria set by each of the five *Daubert* standards. The MNB-RIM consists of individual instruments that are well established and widely used by a large number of psychologists. The MNB-RIM carefully categorizes each test into a cognitive domain based upon the factor structure of the battery, which is consistent with the suggestions of Larrabee (2000). The assignment of scores to a domain prevents redundancy and produces a simplified battery that retains a high level of internal consistency. The MNB-RIM takes approximately 3 h to complete the MNB-RIM, while other batteries can take upwards of 8 h to administer. As a result, examinees are less likely to experience fatigue, which we believe is a common cause of invalid results. The literature also supports the external validity of the MNB-RIM, with improved detection of both the existence of impairments caused by a variety of neuropathologies (e.g., TBI), but the severity of the deficits as well. Several peer-reviewed studies of the MNB-RIM have found it to be more strongly correlated with a variety of TBI biomarkers (Meyers, Galinsky, & Volbrecht 1999a; Meyers & Rohling, 2004, 2009; Rohling et al. 2003b).

The MNB-RIM has an acceptable sensitivity and specificity to mild TBI. The battery's reliability was estimated to be .86 (Meyers & Rohling, 2004). The MNB-RIM has been found to be associated with a host of real life activities, such as driving (Meyers et al. 1999a, b) and correlates well with CT/MRI data (Meyers & Rohling 2009).

Another standard of *Daubert* that is satisfied by the MNB-RIM is control over the methods and use of the battery. There are detailed instructions in the user manual regarding the process of administering the battery, which when followed likely increases the test battery's reliability. General acceptance of the MNB-RIM has been exhibited through clinicians' widespread use of the battery, as well as the extensive peer-reviewed literature that has demonstrated its psychometric properties. These conditions also satisfy the Frye standards,

which are often used in place of the *Daubert* standards in certain jurisdictions.

MNB and/or RIM Criticisms

Experts who use MNB-RIM results and interpretations in trials of *Daubert* admissibility standards should be informed of published criticisms regarding the battery. Awareness of such criticisms will help practitioners to adequately address these claims during testimony. Lezak, Howieson, Bigler, & Tranel (2012) discuss potential problems with the MNB-RIM, noting:

A more serious problem with this kind of approach is that it detracts from appropriate neuropsychological assessment by fostering an overreliance on test scores rather than clinical decisions based on patients as persons, their history and their clinical presentation. Moreover, a set of test scores converted into a graphed "profile" is of limited-if any-benefit for the important social and humanitarian goals of patient care, rehabilitation, counseling, and community reintegration (pp. 747–748).

The critique of Lezak et al. (2012) erroneously simplifies the methods used by practitioners of the MNB-RIM. Qualified clinicians do not solely depend on the MNB-RIM as criteria for diagnosis. Medical history, face-to-face communication with the patient, as well as relevant imaging and neurological information are all taken into account before a diagnosis is made. The MNB-RIM is not an exclusive method used for diagnosis; rather, it contributes to the diagnosis. This careful consideration of all relevant data should be the standard of good clinical judgment and effective clinical practices.

Two critiques related to the RIM as an interpretive method for flexible batteries have been published: one by Palmer, Appelbaum, & Heaton (2004) and another by Willson & Reynolds (2004). These critiques, collectively, focus on eight concerns regarding the RIM. A rebuttal of these critiques was submitted by Rohling et al. (2004). Psychometric theory, as well as various analyses of data, has established the RIM's ability to distinguish scores obtained from normal individuals, as well as scores obtained from patients suffering from various pathologies, such as TBI. Rohling et al. concluded that clinicians should adopt the RIM as a method of reducing and avoiding errors in interpretation common in clinical practice. These interpretation errors are outlined in (Dawes et al. 1989). It is also worth noting that despite the criticisms of Palmer et al., these same authors published a similar method to the RIM examining the stability and course of deficits associated with schizophrenia. It is ironic that they are critical of the RIM, yet chose to use the method for their analyses (Heaton, Gladsjo, Palmer, Kick, Marcotte, and Jeste 2001).

References

- American Educational Research Association, Psychological Association, & National Council on Measurement in Education. (2014). *Standards for educational and psychological testing*. Washington, DC: American Educational Research Association.
- Amicus Brief submitted on behalf of the American Academy of Clinical Neuropsychology. (2007). in *Support of Plaintiff in Baxter v. Temple*, 949 A.2d 167 (N.H. May 20, 2008). Available at <https://www.courtlistener.com/nh/8Ucx/baxter-v-temple/>
- Benton, A., Savin, A. B., de Hamsher, K. S., Varney, N. R., & Spreen, O. (1994). *Contributions to neuropsychological assessment: A clinical manual* (2nd ed.). New York: Oxford University Press.
- Brandt, J., Rubinsky, E., & Larson, G. (1985). Uncovering malingered amnesia. *Annals of the New York Academy of Sciences*, 44, 502–503.
- Bush, S. S. (2011). Meyers neuropsychological battery (MNB). In J. S. Kreutzer, J. Deluca, & B. Caplan (Eds.), *Encyclopedia of clinical neuropsychology*. New York: Springer.
- Bush, S. S., Ruff, R. M., Tröster, A. I., Barth, J. T., Koffler, S. P., Pliskin, N. H., ... Silver, C. H. (2005). Symptom validity assessment: Practice issues and medical necessity NAN policy & planning committee. *Archives of Clinical Neuropsychology*, 20, 419–426.
- Butcher, J. N., Dahlstrom, W. G., Graham, J. R., Tellegen, A., & Kaemmer, B. (1989). *The Minnesota multiphasic personality inventory-2 (MMPI-2): Manual for administration and scoring*. Minneapolis: University of Minnesota Press.
- Carone, A. C., & Bush, S. S. (2013). *Mild traumatic brain injury: Symptom validity assessment and malingering*. New York: Springer.
- Constantinou, M., Bauer, L., Ashendorf, L., Fisher, J. M., & McCaffrey, R. J. (2005). Is poor performance on recognition memory effort measures indicative of generalized poor performance on neuropsychological tests? *Archives of Clinical Neuropsychology*, 20, 191–198.
- Daubert v. Merrell Dow Pharmaceuticals, Inc. (1993). 509 U.S. 579, 589
- Dawes, R. M. (1979). The robust beauty of improper linear models in decision making. *American Psychologist*, 34, 571–582.
- Dawes, R. M., & Corrigan, B. (1974). Linear models in decision making. *Psychological Bulletin*, 81, 95–106.
- Dawes, R., Faust, D., & Meehl, P. (1989). Clinical versus actuarial judgment. *Science*, 243(4899), 1668–1674.
- Demakis, G. J. (2004). Frontal lobe damage and tests of executive processing: A meta-analysis of the category test, stroop test, and trail-making test. *Journal of Clinical and Experimental Neuropsychology*, 26(3), 441–450.
- Dikmen, S. S., Machamer, J. E., Winn, H. R., & Temkin, N. R. (1995). Neuropsychological outcome at 1-year post head injury. *Neuropsychology*, 9, 80–90.
- Glass, G. V., McGraw, B., & Smith, M. L. (1981). *Meta-analysis in social research*. Beverly Hills: Sage.
- Goodwin, J. (2010). *Research in psychology methods and designs* (6th ed., p. 191). Hoboken: Wiley.
- Green, P. (2003). *Word memory test*. Edmonton: Green Publishing.
- Green, P. (2004). *Green's medical symptom validity test (MSVT) for Microsoft windows. User's manual*. Edmonton: Green's Publishing.
- Green, P. (2007). The pervasive influence of effort on neuropsychological tests. *Physical Medicine Rehabilitation Clinics of North America*, 18(1), 43–68.
- Green, P., Flaro, L., & Courtney, J. (2009). Examining false positives on the word memory test in adults with mild traumatic brain injury. *Brain Injury*, 23(9), 741–750.
- Green, P., Rohling, M. L., Lees-Haley, P., & Allen, L. M. (2001). Effort accounts for more neuropsychological test variance than does ability in litigant sample. *Brain Injury*, 15, 1045–1060.
- Greiffenstein, M. F. (2008). Basics of forensic neuropsychology. In J. E. Morgan & J. Richter (Eds.), *Forensic neuropsychology* (pp. 3–16). New York: Guilford Press.
- Greiffenstein, M. F., & Kaufmann, P. M. (2012). Neuropsychology and law: Principles of productive attorney-neuropsychologist relations. In G. J. Larrabee (Ed.), *Forensic neuropsychology: A scientific approach* (2nd ed.). New York: Oxford University Press.
- Heaton, R. K., Chelune, G. J., Talley, J. L., Kay, G. G., & Curtiss, G. (1993). Wisconsin Card Sorting Test manual. Odessa, FL: Psychological Assessment Resources.
- Heaton, R. K., Gladsjo, J. A., Palmer, B. W., Kick, J., Marcotte, T. D., & Jeste, D. V. (2001). Stability and course of neuropsychological deficits in schizophrenia. *Archives of General Psychiatry*, 58, 24–32.
- Heaton, R. K., Grant, I., & Matthews, C. G. (1991). *Comprehensive norms for an expanded Halstead-Reitan Battery*. Lutz: Psychological Assessment Resources.
- Heaton, R. K., Miller, W., Taylor, M. J., & Grant, I. (2004). *Revised comprehensive norms for an expanded Halstead-Reitan Battery: Demographically adjusted neuropsychological norms for African American and Caucasian adults*. Lutz: Psychological Assessment Resources.
- Heilbronner, R. L., Sweet, J. J., Morgan, J. E., Larrabee, G. J., Millis, S. R., & Conference participants. (2009). American academy of clinical neuropsychology consensus conference statement on the neuropsychological assessment of effort, response bias, and malingering. *The Clinical Neuropsychologist*, 23(7), 1093–1129.
- Ingraham, L. J., & Aikken, C. B. (1996). An empirical approach to determining criteria for abnormality in test batteries with multiple measures. *Neuropsychology*, 10, 120–124.
- Ivnik, R. J., Malec, J. F., Tangalos, E. G., Peterson, R. C., Kokmen, E., & Kurland, L. T. (1990). The auditory verbal learning test (AVLT): Norms for ages 55 years and older. *Psychological Assessment*, 2(3), 304–312.
- Kaplan, E., Goodglass, H., & Weintraub, S. (1983). *The Boston naming test*. Philadelphia: Lea & Febiger.
- Kaufmann, P. M. (2012). Admissibility of expert opinions based on neuropsychological evidence. In G. J. Larrabee (Ed.), *Forensic neuropsychology: A scientific approach* (2nd ed.). New York: Oxford University Press.
- Kaufmann, P. M., & Greiffenstein, M. F. (2013). In M. L. Mattingly & E. Rinehardt (Eds.), *Forensic neuropsychology: Training, scope of practice, and quality control*. *National Academy of Neuropsychology Bulletin*, 27(1), 11–15.
- Kieman, R. J., & Matthews, C. G. (1976). Impairment index versus T-score averaging in neuropsychological assessment. *Journal of Consulting and Clinical Psychology*, 44, 951–957.
- Kozel, J. J., & Meyers, J. E. (1998). A cross-validation study of the Victoria revision of the category test. *Archives of Clinical Neuropsychology*, 13, 327–332.
- Labreche, T. M. (1983). *The Victoria revision of the Halstead Category Test*. Unpublished doctoral dissertation. University of Victoria.
- Larrabee, G. J. (2000). Association between IQ and neuropsychological test performance: Commentary on Tremont, Hoffman, Scott, and Adams (1998). *The Clinical Neuropsychologist*, 14, 139–145.
- Larrabee, G. J. (2008). Aggregation across multiple indicators improves the detection of malingering: Relationship to likelihood ratios. *The Clinical Neuropsychologist*, 22, 666–679.
- Larrabee, G. J. (2012a). A scientific approach to forensic neuropsychology. In G. J. Larrabee (Ed.), *Forensic neuropsychology: A scientific approach* (2nd ed., pp. 3–22). New York: Oxford University Press.
- Larrabee, G. J. (2012b). Performance validity and symptom validity in neuropsychological assessment. *Journal of the International Neuropsychological Society*, 18(4), 625–630.
- Larrabee, G. J., Millis, S. R., & Meyers, J. E. (2008). Sensitivity to brain dysfunction of the Halstead Reitan vs. an ability-focused neuropsychological battery. *The Clinical Neuropsychologist*, 22(5), 813–825.
- Larrabee, G. J., Millis, S. R., & Meyers, J. E. (2009). 40 plus or minus 10: A new magical number: Reply to Russell. *The Clinical Neuropsychologist*, 23(5), 841–849.

- Leonberger, F. T., Nicks, S. D., Larrabee, G. J., & Goldfader, P. R. (1992). Factor structure of the Wechsler memory scale-revised within a comprehensive neuropsychological battery. *Neuropsychology*, 6(3), 239–249.
- Lezak, M. D. (1995). *Neuropsychological assessment*. New York: Oxford University Press.
- Lezak, M. D., Howieson, D. B., Bigler, E. D., & Tranel, D. (2012). *Neuropsychological assessment* (5th ed., pp. 747–748). New York: Oxford University Press.
- Lezak, M. D., Howieson, D. B., & Loring, D. (2004). *Neuropsychological assessment* (4th ed., p. 636). New York: Oxford University Press.
- Maroco, J., Silva, D., Rodriguez, A., Guerreiro, M., Santana, I., & de Mendonça, A. (2011). Data mining methods in the prediction of dementia: A real-data comparison of the accuracy, sensitivity and specificity of linear discriminant analysis, logistic regression, neural networks, support vector machines, classification trees and random forests. *BMC Research Notes*, 4, 299–313.
- Meehl, P. E. (1954). Clinical versus statistical prediction: A theoretical analysis and a review of the evidence. Minneapolis, MN: University of Minnesota.
- Meyers, J. E. (2014). *Manual for the Meyers Neuropsychological Battery (MNB)* (Electronic). Retrieved on 07/14/2015 from <http://www.meyersneuropsychological.com/>.
- Meyers, J. E., & Diep, A. (2000). Assessment of malingering in chronic pain patients using neuropsychological tests. *Applied Neuropsychology*, 7, 133–139.
- Meyers, J. E., Galinsky, A., & Volbrecht, M. E. (1999a). Malingering and mild brain injury: How low is too low. *Applied Neuropsychology*, 6, 208–216.
- Meyers, J. E., & Meyers, K. R. (1995a). *Rey complex figure test and recognition trial: Professional manual*. Odessa: Psychological Assessment Resources.
- Meyers, J. E., & Meyers, K. R. (1995b). The Rey complex figure and recognition trial under four different administration procedures. *The Clinical Neuropsychologist*, 9, 65–67.
- Meyers, J. E., Miller, R. M., Haws, N. A., Murphy-Taft, J. L., Curtis, T. D., Rupp, Z. W., ... Thompson, L. M. (2013a). An adaptation of the MMPI-2 Meyers index for the MMPI-2-RF. *Applied Neuropsychology Adult*, 21(2), 148–154.
- Meyers, J. E., Miller, R. M., Lisa, M., Thompson, L. M., Scalese, A. M., Allred, B. C., ... Lee, A. J.-H. (2014a). Using likelihood ratios to detect invalid performance with performance validity measures. *Archives of Clinical Neuropsychology*, 29, 224–235.
- Meyers, J. E., Miller, R. M., & Tuita, A. R. (2014b). Using pattern analysis matching to differentiate TBI and PTSD in a military sample. *Applied Neuropsychology Adult*, 21(1), 60–68.
- Meyers, J. E., Millis, S. R., & Volkert, K. T. (2002a). A validity index for the MMPI-2. *Archives of Clinical Neuropsychology*, 17(2), 157–169.
- Meyers, J. E., Reinsch-Boothby, L., Miller, R. M., Rohling, M. L., & Axelrod, B. N. (2011a). Does the source of a forensic referral affect neuropsychological test performance on a standardized battery of tests? *The Clinical Neuropsychologist*, 25, 477–487.
- Meyers, J. E., Roberts, R. J., Bayless, J. D., Volkert, K. T., & Evitts, P. E. (2002b). Dichotic listening: Expanded norms and clinical application. *Archives of Clinical Neuropsychology*, 17(1), 79–90.
- Meyers, J. E., & Rohling, M. L. (2004). Validation of the Meyers short battery on mild TBI patients. *Archives of Clinical Neuropsychology*, 19, 637–651.
- Meyers, J. E., & Rohling, M. L. (2009). CT and MRI correlations with neuropsychological tests. *Applied Neuropsychology*, 16(4), 237–253.
- Meyers, J. E., & Volbrecht, M. E. (2003). A validation of multiple malingering detection methods in a large clinical sample. *Archives of Clinical Neuropsychology*, 18(3), 261–276.
- Meyers, J. E., Volbrecht, M. E., Axelrod, B. N., & Reinsch-Boothby, L. (2011b). Embedded symptom validity tests and overall neuropsychological test performance. *Archives of Clinical Neuropsychology*, 26, 8–15.
- Meyers, J. E., Volbrecht, M. E., & Kaster-Bundgaard, J. (1999b). Driving is more than pedal pushing. *Applied Neuropsychology*, 6, 154–164.
- Meyers, J. E., Volkert, K. T., & Diep, A. (2000). Sentence repetition test: Updated norms and clinical utility. *Applied Neuropsychology*, 7, 154–159.
- Meyers, J. E., Zellinger, M. M., Kockler, T., Wagner, M., & Miller, R. M. (2013b). A Validated seven-subtest short form for the WAIS-IV. *Applied Neuropsychology*, 20, 249–256.
- Miller, J. B., Fichtenberg, N. L., & Millis, S. R. (2010). Diagnostic efficiency of an ability-focused battery. *The Clinical Neuropsychologist*, 24(4), 678–688.
- Miller, L. S., & Rohling, M. L. (1998). *Statistical interpretive methods for difficult neuropsychological test data*. Poster presented at the 18th annual conference of the National Academy of Neuropsychology in Washington, DC.
- Miller, L. S., & Rohling, M. L. (2001). A statistical interpretive method for neuropsychological data. *Neuropsychology Review*, 11(3), 143–169.
- Mitrushina, M., Boone, K. B., Razani, J., & D'Elia, L. F. (2005). *Handbook of normative data for neuropsychological assessment* (2nd ed.). New York: Oxford University Press.
- Mitrushina, M., Boone, K. B., & D'Elia, L. F. (1999). *Handbook of normative data for neuropsychological assessment*. New York: Oxford University Press.
- Palmer, B. W., Appelbaum, M. I., & Heaton, R. K. (2004). Rohling's interpretive method and inherent limitations on the flexibility of flexible batteries. *Neuropsychology Review*, 14(3), 171–176.
- Perrine, K. (1993). Differential aspects of conceptual processing in the category test and the Wisconsin card sort test. *Journal of Clinical and Experimental Neuropsychology*, 15(4), 461–473.
- Pilgrim, B., Meyers, J. E., Bayless, J., & Whetstone, M. (1999). Validity of the ward seven-subtest WAIS-III short form in a neuropsychological population. *Applied Neuropsychology*, 6, 243–246.
- Pohar, M., Blas, M., & Turk, S. (2004). Comparison of logistic regression and linear discriminant analysis: A simulation study. *Metodološki zvezki*, 1, 143–161.
- Rabin, L. A., Barr, W. B., & Burton, L. A. (2005). Assessment practices of clinical neuropsychologists in the United States and Canada: A survey of INS, NAN, and APA Division 40 members. *Archives of Clinical Neuropsychology*, 20(1), 33–65.
- Read, D. E., & Spreen, O. (1986). Normative data in older adults for selected neuropsychological tests. In O. Spreen & Strauss (Eds.; 1998), *A compendium of neuropsychological tests* (2nd ed.; p. 181). Unpublished data: University of Victoria.
- Reitan, R. M., & Wolfson, D. (1985). *The Halstead-Reitan neuropsychological test battery: Theory and interpretation*. Tucson: Neuropsychology Press.
- Reitan, R. M., & Wolfson, D. (1988). *Traumatic brain injury volume II: Recovery and rehabilitation*. Tucson: Neuropsychology Press.
- Reitan, R. M., & Wolfson, D. (1997). Emotional disturbances and their interaction with neuropsychological deficits. *Neuropsychology Review*, 7(1), 3–19.
- Roberts, M., Persinger, M., Grote, C., Evertowski, L., Springer, J., Tuten, T., ... Baglio, C. (1994). The dichotic listening test: Preliminary observations in American and Canadian samples. *Applied Neuropsychology*, 1, 45–56.
- Rohling, M. L., Langhinrichsen-Rohling, J., & Miller, L. S. (2003a). Statistical methods for determining malingering: Rohling's interpretive method. In R. Franklin (Ed.), *Prediction in forensic and neuropsychology: Sound statistical practices* (pp. 171–207). Mahwah: Lawrence Erlbaum Associates.
- Rohling, M. L., Meyers, J. E., & Millis, S. R. (2003b). Neuropsychological impairment following traumatic brain injury:

- A dose–response analysis. *The Clinical Neuropsychologist*, 17, 289–302.
- Rohling, M. L., Miller, R. M., Axelrod, B. N., Wall, J. H., Lee, A. J., Kinikini, D. T. (2015). Is co-norming required? *Archives of Clinical Neuropsychology*. Published online. Accessed 07/14/2015.
- Rohling, M. L., Miller, L. S., Langhinrichsen-Rohling, J. (1998). *Rohling's evaluation method for interpreting neuropsychological data (REMIND)*. Poster presented at the 28th annual conference of the International Neuropsychological Society in Honolulu, HI.
- Rohling, M. L., Miller, L. S., & Langhinrichsen-Rohling, J. (2004). Rohling's interpretive method for neuropsychological case data: A response to critics. *Neuropsychology Review*, 14(3), 155–169.
- Rohling, M. L., Williamson, D. J., Miller, L. S., & Adams, R. L. (2003c). Using the Halstead-Reitan Battery to diagnose brain damage: A comparison of the predictive power of traditional techniques to Rohling's interpretive method. *The Clinical Neuropsychologist*, 17, 531–543.
- Rosenthal, R. (1984). *Meta-analytic procedures for social research*. Beverly Hills: Sage.
- Rosenthal, R., Rosnow, R. L., & Rubin, D. B. (2000). *Contrasts and effect sizes in behavioral research*. New York: Cambridge University Press.
- Russell, E. W. (2012). *Scientific foundations of neuropsychological assessment with application to forensic evaluation* (pp. 297–298). Waltham: Elsevier.
- Russell, E. W., Neuringer, C., & Goldstein, G. (1970). *Assessment of brain damage: A neuropsychological key approach*. New York: Wiley.
- Russell, E. W., & Starkey, R. I. (2001). *Halstead Russell neuropsychological evaluation system-revised (HRNES-R)*. Los Angeles: Western Psychological Services.
- Sicoly, F. (1989). Computer-aided decisions in human services: Expert systems and multivariate models. *Computers in Human Behavior*, 5, 47–60.
- Spree, O., & Strauss, E. M. (1991). *A compendium of neuropsychological tests: Administration, norms, and commentary*. New York: Oxford University Press.
- Spree, O., & Strauss, E. M. (1998). *A compendium of neuropsychological tests: Administration, norms, and commentary* (2nd ed.). New York: Oxford University Press.
- Stern, R. A., & White, T. (2003). *Neuropsychological assessment battery: Administration, scoring, and interpretation manual*. Lutz: Psychological Assessment Resources.
- Stevens, A., Friedel, E., Mehren, G., & Merten, T. (2008). Malingering and uncooperativeness in psychiatric and psychological assessment: Prevalence and effects in a German sample of claimants. *Psychiatry Research*, 157, 191–200.
- Strauss, E., Sherman, E. M., & Spreen, O. (2006). *A compendium of neuropsychological tests: Administration, norms, and commentary* (3rd ed.). New York: Oxford University Press.
- Sweet, J. J., King, J. H., Malina, A. C., Bergman, M. A., & Simmons, A. (2002). Documenting the prominence of forensic neuropsychology at national meetings and relevant professional journals, from 1990 to 2000. *The Clinical Neuropsychologist*, 16, 481–494.
- Sweet, J. J., & Meyer, D. G. (2011). Trends in forensic practice and research. In G. J. Larrabee (Ed.), *Forensic neuropsychology: A scientific approach* (pp. 501–516). New York: Oxford University Press.
- Sweet, J. J., Meyers, D. G., Nelson, N. W., & Moberg, P. J. (2011). The TCN/AACN 2010 salary survey: Professional practices, beliefs, and incomes if U.S. neuropsychologists. *The Clinical Neuropsychologist*, 25(1), 12–61.
- Tellegen, A., & Ben-Porath, Y. S. (2008). *MMPI-2-RF (Minnesota multiphasic personality inventory-2 restructured form): Technical manual*. Minneapolis: University of Minnesota Press.
- Tombaugh, T. N. (1996). *TOMM. Test of memory malingering*. New York: Multi-Health Systems Inc.
- Tversky, A., & Kahneman, C. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157), 1124–1131.
- Ward, L. C. (1990). Prediction of verbal, performance, and full scale IQs from seven subtests of the WAIS-R. *Journal of Clinical Psychology*, 46(4), 436–440.
- Wechsler, D. (1997). *Manual for the Wechsler adult intelligence scale* (3rd ed.). San Antonio: The Psychological Corporation.
- Wechsler, D. (2003). *Manual for the Wechsler intelligence scale for children* (4th ed.). San Antonio: The Psychological Corporation.
- Wechsler, D. (2008). *Manual for the Wechsler adult intelligence scale* (4th ed.). San Antonio: Pearson.
- Wedding, D., & Faust, D. (1989). Clinical judgment and decision making in neuropsychology. *Archives of Clinical Neuropsychology*, 4, 233–265.
- Williams, J. M. (1997). The prediction of premorbid memory ability. *Archives of Clinical Neuropsychology*, 12, 745–756.
- Willson, V. L., & Reynolds, C. R. (2004). A critique of Miller and Rohling's statistical interpretive method for neuropsychological test data. *Neuropsychology Review*, 14(3), 177–181.